

Soft Rateless Optical Communication with a Mobile Camera Receiver

In collaboration with
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Introduction

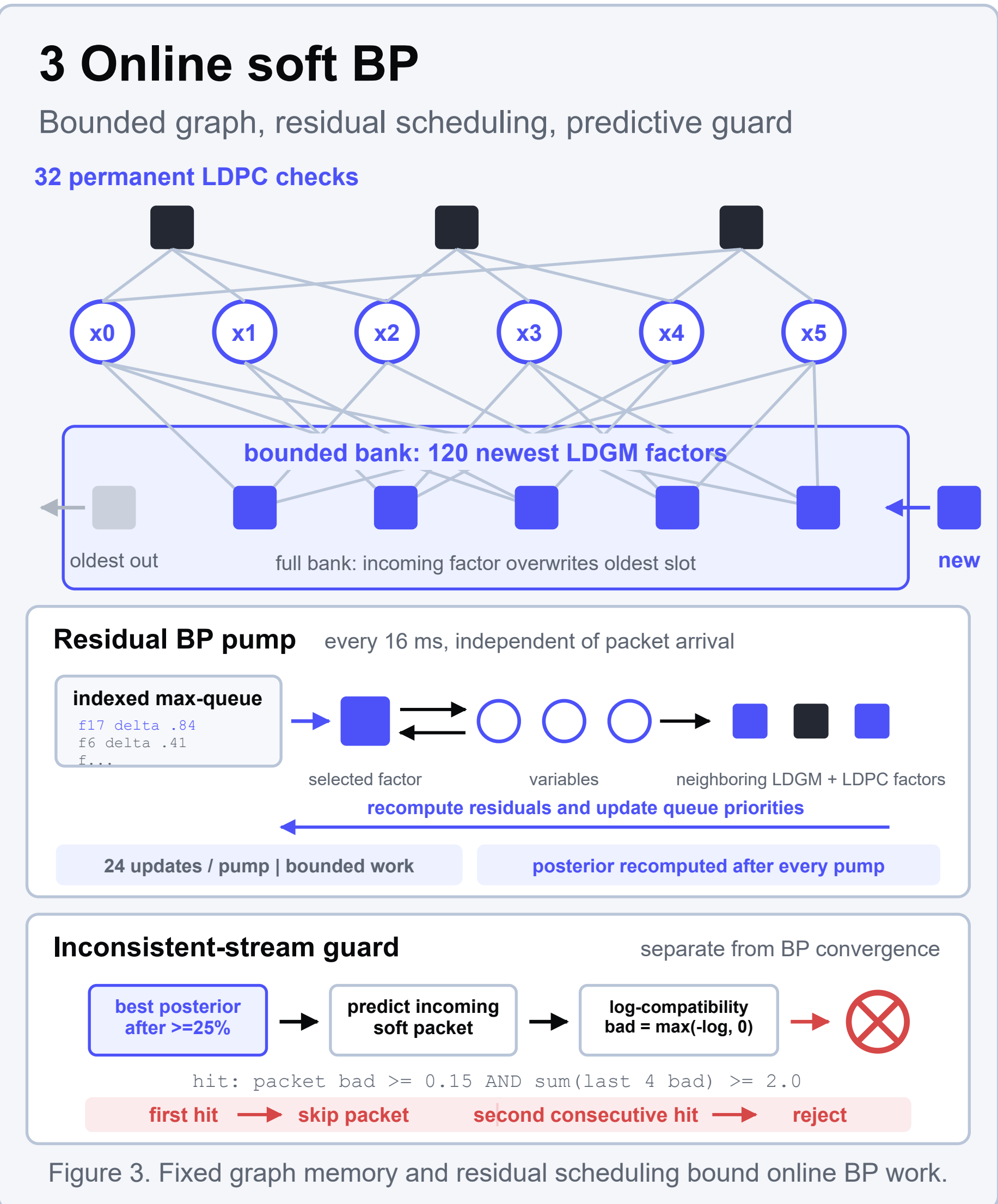
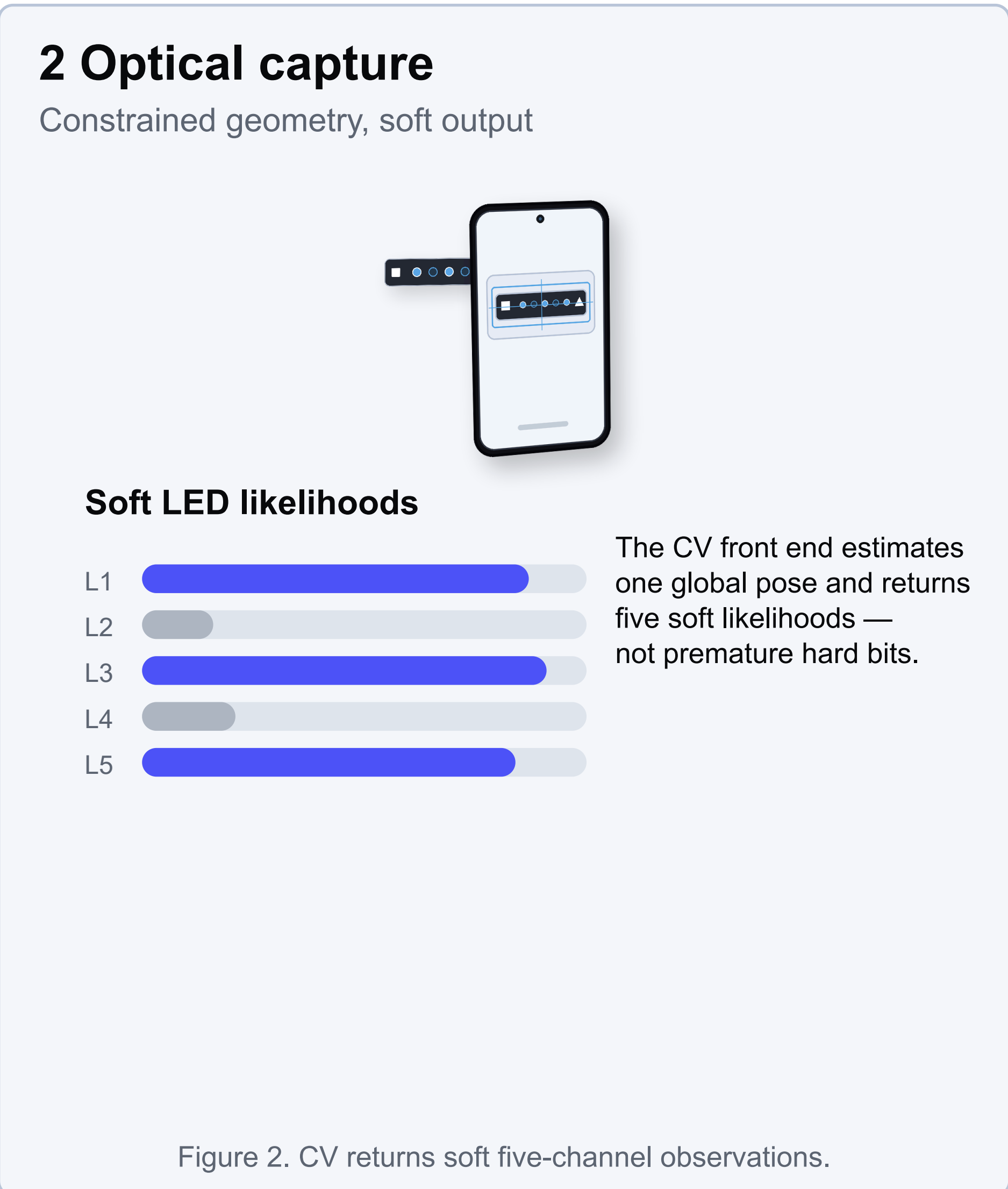
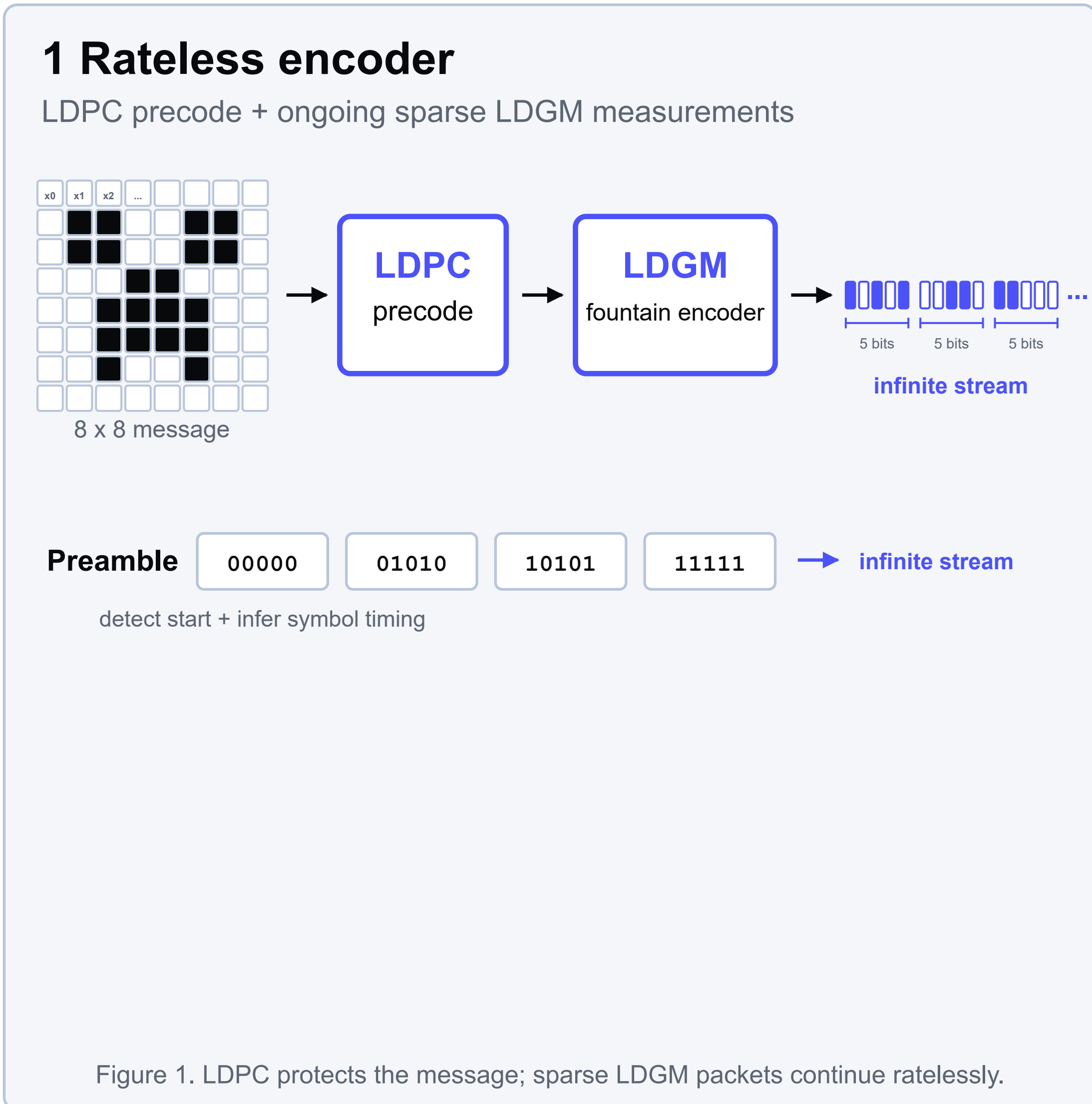
A phone camera observes five LEDs as an asynchronous, noisy optical channel. Hard frame-by-frame decoding was too fragile: a few uncertain observations could produce a confidently wrong message. The project therefore separates computer vision from communication decoding. CV produces timestamped soft LED evidence; a rateless receiver synchronizes the stream and accumulates information online.

Core novelty: soft rateless reception with persistent online belief propagation.

Objectives

- Detect a constrained start-marker / five-LED / end-marker pattern in real time.
- Infer symbol timing from a known preamble and sample soft LED observations.
- Recover an 8 x 8 binary message without requiring a fixed packet count.
- Preserve useful progress through short tracking gaps and reject inconsistent streams.
- Integrate the complete receiver workflow into an Android prototype.

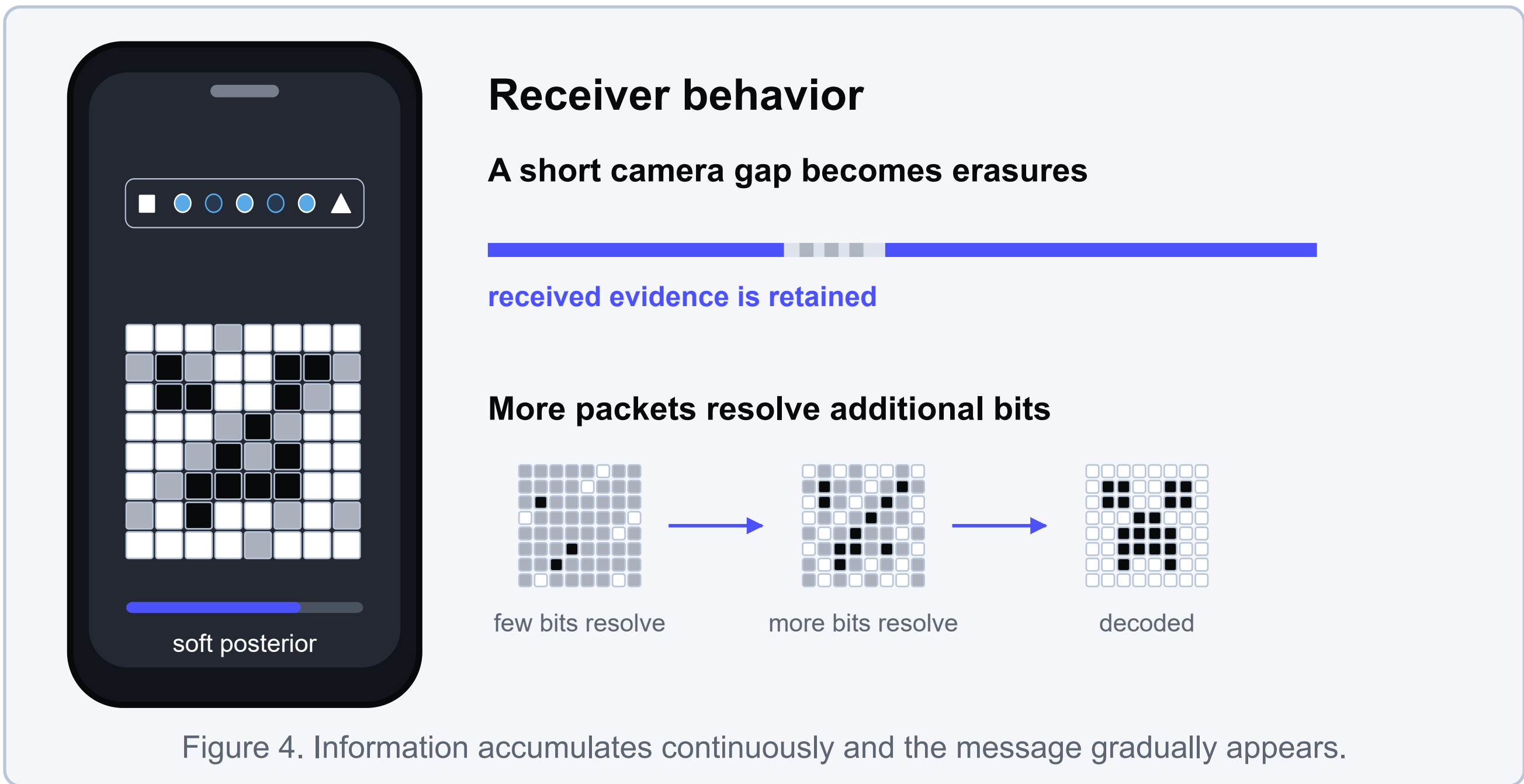
Process & Methods



Results

Working end-to-end prototype

- Android CameraX + Jetpack Compose receiver with acquisition, decoding progress, cancellation, and final message view.
- Browser-based virtual transmitter with the marker pattern, five LED channels, editable binary message, and selectable rates.
- Modular interface between CV and protocol: the decoder consumes timestamped soft observations and erasures, not camera frames.
- Preamble-based timing estimation, timing-window packet sampling, and online factor-graph decoding are integrated in one workflow.
- Rateless reception retains accumulated evidence after short gaps; better CV makes convergence faster without changing the protocol.



Conclusions

The project established a proprietary architecture for a mobile optical receiver: constrained CV supplies soft evidence, while a Raptor-like LDPC+LDGM stream and online BP turn unreliable five-bit observations into accumulated information. Compared with hard direct transmission, the receiver is less brittle and does not restart after every short observation gap. The prototype can be connected to a future physical LED device; ongoing work focuses on faster mobile vision, hardware validation, and broader phone tests.

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